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Shahadat Hussain

Research Associate

Affiliation: NYU Abu Dhabi
Education: PhD Academy of Scientific and Innovative Research

shahadat.hussain@nyu.edu

Research Areas: Additive Manufacturing, Metamaterials, TPMS Lattices, Interpenetrating Phase Composites (IPCs), Hybrid Manufacturing, Smart Materials, NiTi, Shape Memory Alloys, 4D Printing, Energy Absorption, and Investment Casting (Gypsum-based)

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Shahadat Hussain is a Research Associate in the Division of Engineering at New York University Abu Dhabi. He holds a Doctor of Philosophy in Materials Science and Technology from the Academy of Scientific and Innovative Research (AcSIR) at CSIR-Advanced Materials and Processes Research Institute, Bhopal, India, and a Bachelor of Engineering in Mechanical Engineering from UIT-RGPV, Bhopal. With over a decade of research experience spanning institutions in India and the United Arab Emirates, Hussain has developed deep expertise across the full spectrum of advanced materials research – from computational design and additive manufacturing to experimental characterization and mechanical testing.

Prior to joining NYU Abu Dhabi, he held a postdoctoral fellowship at Khalifa University, Abu Dhabi, where he conducted research on NiTi shape memory alloy TPMS lattice structures fabricated via Laser Powder Bed Fusion (LPBF), investigating the relationships between process parameters, microstructural evolution, and phase transformation behavior in these architected systems.

Summary of Research

Hussain's research centers on the development of advanced metamaterial structures for lightweight, structural, vibration-absorbing, and damping applications. His work employs Fused Filament Fabrication (FFF) and Stereolithography (SLA) to fabricate Triply Periodic Minimal Surface (TPMS) geometries – including primitive and gyroid topologies – from carbon fiber reinforced polymer filaments, with extensive compression and four-point bending testing carried out to characterize their mechanical response.

A significant thrust of his current research involves the creation of Interpenetrating Phase Composites (IPCs), in which 3D-printed TPMS lattice structures are impregnated with various resins to produce multiphase architected materials with enhanced structural and damping performance. He has further extended this approach to hybrid manufacturing, where additively manufactured polymer TPMS patterns are used as sacrificial templates for aluminum alloy casting, producing metal TPMS beams that are subsequently impregnated with resins – combining the geometric freedom of additive manufacturing with the mechanical advantages of metal casting.